Python Programming - Unit Wise 5M and 10M Important Questions

UNIT I

10 Marks Questions

1. Explain the Program Development Cycle with a neat diagram.
2. Explain Python data types with suitable examples.
3. Explain different operators in Python with examples.
4. Explain decision-making statements with examples.
5. Explain looping statements (while and for loops) with examples.
6. Explain variables, type conversions, expressions, and keyboard input in Python.

5 Marks Questions

1. What are the phases of the Program Development Cycle?
2. Explain the print() function with examples.
3. Explain comments in Python.
4. Differentiate between variables and constants.
5. Explain type conversion with examples.
6. Explain logical operators.
7. Explain Boolean expressions.
8. Explain nested decisions.
9. Explain running totals.
10. Explain input validation.
11. Explain nested loops.

UNIT II

10 Marks Questions

1. Explain string operations and string methods with examples.
2. Explain Lists in Python and various list operations.
3. Explain Tuples and compare them with Lists.
4. Explain Dictionaries and dictionary methods with examples.
5. Explain Sets and set operations with examples.
6. Explain String Formatting techniques in Python.

5 Marks Questions

1. Explain basic string operations.
2. Explain string methods.
3. Define List and explain its characteristics.
4. Define Tuple and explain tuple() function.
5. Differentiate between List and Tuple.
6. Explain dictionary creation and access.
7. Explain dictionary methods.
8. Explain set creation.
9. Explain set methods.
10. Explain simple set comprehensions.

UNIT III

10 Marks Questions

1. Explain functions in Python with syntax and examples.
2. Explain parameter passing mechanisms in Python.
3. Differentiate between local and global variables.
4. Explain default arguments and keyword arguments with examples.
5. Explain lambda functions with suitable examples.
6. Explain modules and packages in Python.
7. Explain recursion with suitable examples.
8. Explain top-down design approach in problem solving.

5 Marks Questions

1. Define a function.
2. Explain parameters and arguments.
3. Explain return statement.
4. What are local variables?
5. What are global variables?
6. Explain default arguments.
7. Explain keyword arguments.
8. What is a lambda function?
9. Explain import statement.
10. Explain from-import statement.
11. What is a package?
12. Explain recursion.

UNIT IV

10 Marks Questions

1. Explain Object-Oriented Programming concepts in Python.
2. Explain classes, objects, and instances with examples.
3. Explain constructors and destructors with examples.
4. Explain inheritance with suitable examples.
5. Explain polymorphism in Python.
6. Explain operator overloading with examples.
7. Explain file handling operations in Python.
8. Explain read(), readline(), and readlines() with examples.
9. Explain write() and writelines() with examples.
10. Explain seek() and file pointer manipulation.

5 Marks Questions

1. Define class and object.
2. Explain instances.
3. Explain constructor.
4. Explain destructor.
5. Explain class attributes.
6. Explain inheritance.
7. What is polymorphism?
8. What is operator overloading?
9. Explain read() function.
10. Explain readline() function.
11. Explain readlines() function.
12. Explain write() function.
13. Explain writelines() function.
14. Explain seek() function.
15. Explain log file handling.

UNIT V

10 Marks Questions

1. Explain different types of errors and exceptions in Python.
2. Explain exception handling using try, except, else, and finally blocks.
3. Explain raising exceptions with examples.
4. Explain user-defined exceptions with examples.
5. Explain clean-up actions and redefined clean-up actions.
6. Explain GUI-based programs and their advantages over terminal-based programs.
7. Develop a simple GUI application using Python.

5 Marks Questions

1. What is a Syntax Error?
2. What is an Exception?
3. Explain try and except blocks.
4. Explain finally block.
5. Explain raise statement.
6. What are user-defined exceptions?
7. What are clean-up actions?
8. Differentiate between syntax errors and exceptions.
9. Explain terminal-based programs.
10. Explain GUI-based programs.
11. List advantages of GUI applications.